Access Instructions

Login Required?

No. BreakCheck has no login, sign-up, or user accounts. There are no test credentials to share, and judges do not need to create an account.

Option 1: Live App (quickest)

1. Open ***https://breakcheck-live.vercel.app/***
2. Click the **Orders API** project.
3. Start on **Overview**, then use the tabs: Versions, Diff, Impact, Tasks, Activity, Consumers.

Note: on the hosted site the Activity page uses a 5-second polling fallback (shown with an amber dot), because serverless hosting cannot hold a change stream open. The true live change-stream feed (green dot) runs locally.

Option 2: Try It Yourself (5 steps)

1. Go to **Versions** and upload orders-v1.yaml, then orders-v2.yaml (sample files are in the samples/ folder of the GitHub repo). Give them the labels v1 and v2.
2. Go to **Impact**, choose **From = v1**, **To = v2**, and click **Who breaks?**
3. See the **Do not ship** verdict and the impact matrix. Analytics Job is not flagged, because it never reads the removed total field.
4. Open **Tasks** to see migration tasks with fix hints. Tick a checkbox and refresh; it stays saved.
5. Upload orders-v3.yaml as v3 and compare **v2 → v3**. Open **Activity** to see the events.

Try the Failure Cases

- Upload samples/orders-invalid.yaml: a red panel shows the location of each problem.
- Upload orders-v1.yaml twice: the second upload says “Already ingested.”
- Send the same webhook twice (see below): the second is ignored.

Webhook test (PowerShell):

```
powershell
$b = '{"eventId":"judge-1","project":"orders-api","versionLabel":"v-test","spec":"openapi:
3.0.3\ninfo:\n  title: T\n  version: 1\npaths: {}\n"}'
Invoke-RestMethod -Method Post -Uri https://breakcheck-live.vercel.app/api/webhooks/spec
-ContentType "application/json" -Body $b
Invoke-RestMethod -Method Post -Uri https://breakcheck-live.vercel.app/api/webhooks/spec
-ContentType "application/json" -Body $b
```

The first reply says processed; the second says duplicate_ignored. Use a new eventId for each fresh test.

Option 3: Run Locally

1. git clone https://github.com/Harsheel-Kumar-Sahoo/BreakCheck.git then cd BreakCheck and npm install
2. Copy .env.example to .env.local and set MONGODB_URI to your own free MongoDB Atlas (M0) connection string. It must be a replica set for change streams. Set MONGODB_DB=breakcheck.
3. npm run setup-db (creates collections, validators, indexes)
4. npm run seed (creates the Orders API project and 5 sample consumers)
5. npm run dev, then open *http://localhost:3000*

Run the CLI (no database needed):

bash
npx tsx cli/breakcheck.ts compare samples/orders-v1.yaml samples/orders-v2.yaml --consumers samples/consumers.json

Exit code 1 means breaking changes hit consumers; 0 means safe; 2 means bad input.

Run the tests: npm test

Notes

- The hosted app shares one demo database containing only sample data. Please avoid uploading real or sensitive specs.
- No personal passwords or secrets are included in the repository. .env.local is excluded from Git, and each user supplies their own MongoDB connection string.